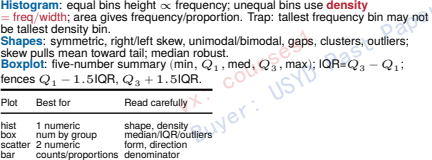


STATS/202 2651 FINAL

Exam map: MCQ + extended answer. Always state H/A/T/P/C, assumptions, statistic, p-value/CI, conclusion in context.

W1 GRAPHICAL SUMMARIES

Data types: unit/ratio, variable/continuous; categorical/nominal/ordinal vs quantitative discrete/continuous; response Y, explanatory X. Study design: population → sampling frame → sample; random sample reduces selection bias; random allocation supports causal claim; blinding/placebo reduces measurement/expectation bias. Association ≠ causation: confounding, reverse causation, selection, non-response, survivorship. Stronger causal evidence: randomized controlled experiment; otherwise adjust stratify and be cautious. Histogram: equal bins height < frequency; unequal bins use density = freq./width; area gives frequency/proportion. Tap: tallest frequency bin may not be tallest density bin. Shapes: symmetric, right/left skew, unimodal/bimodal, gaps, clusters, outliers; skew pulls mean toward tail; median robust. Boxplot: five-number summary (min, Q1, median, Q3, max); IQR = Q3 - Q1; fences Q1 - 1.5IQR, Q3 + 1.5IQR.



Plot	Best for	Read carefully
hist	1 numeric	shape, density
scatter	2 numeric	median/IQR/outliers
bar	counts/proportions	direction



W2 NUMERICAL SUMMARIES

Centre: mean $\bar{x} = \sum x_i/n$ balances; median is 50th pct; mode rarely enough. If add a : mean, medians adds a , SD/IQR unchanged. If multiply by b : centre $\times b$, SD/IQR $\times |b|$, correlation sign may change if one var multiplied by negative. Spread: sample variance $s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$; SD s has original units; RMS deviation from mean uses n denominator.

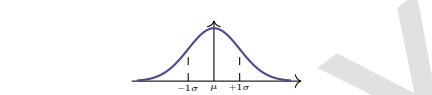
$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$, $z_i = (x_i - \bar{x})/s_x$. Robustness: median/IQR robust to outliers; mean/SD sensitive. For skewed data report median/IQR, for near symmetric report mean/SD. Comparing groups: compare centre + spread + shape + overlap + sample size. Avoid saying 'larger mean/smaller' when typical value is asked. Standard units: $z = (x - \mu)/\sigma$ or $(x - \bar{x})/s$; $z = 2$ means two SD above centre; $z = -2$ means two SD below centre. Transformation: log reduces right skew and turns multiplicative changes into additive differences; if Y log-scale, coefficient b approx 100**b**% change for 1-unit X.

Change	Mean	SD
$x + a$	$\bar{x} + a$	
bx	$b\bar{x}$	$ b s$
standardize 0	$\bar{x} - \bar{x}$	$s/s = 1$
log	geometric focus	ratio scale

Micro-example: data 2, 3, 3, 8, 8; $\bar{x} = 4$, median 3, s large because 8 pulls mean/tail; IQR describes middle half.

W3 NORMAL CURVE + CORR

Normal model: $X \sim N(\mu, \sigma^2)$ in math; R uses mean, sd: `pnorm(x, mu, sd)`, `qnorm(p, mu, sd)`, `dnorm(x, mu, sd)`. $P(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$. R trap: `dnorm` is density height, not probability; interval probability uses `pnorm`; percentile/cutoff uses `qnorm`. Empirical rule: approx 68/95/99.7 within 1/2/3 SD when Normal. Tails are symmetric; upper 10% cutoff $z = \mu + 1.2816\sigma$. Correlation: $r = \frac{s_y}{s_x} \frac{\bar{y} - \bar{y}_0}{\bar{x} - \bar{x}_0}$, $-1 \leq r \leq 1$; measures linear association only; unitless; sensitive to outliers/restriction of range. If $X' = a + bX$, $Y' = c + dY$, then $\text{corr}(X', Y') = \text{sign}(b/d) \text{corr}(X, Y)$. Scatter reading: direction, form, strength, outliers, groups. $r = 0$ means no linear association, not necessarily no relation.



W4 LINEAR MODELS

Least squares line: $\hat{y} = a + bx$ minimizes $\sum e_i^2$; residual $e_i = y_i - \hat{y}_i$; with intercept $\sum e_i = 0$, residual mean 0. $b = \frac{s_y}{s_x} r$, $a = \bar{y} - b\bar{x}$, $\hat{y} = \bar{y} + r \frac{s_y}{s_x} (x - \bar{x})$. RMS error: typical vertical prediction error; in simple regression $s_e \approx s_y \sqrt{1 - r^2}$ (course convention) may use RMS around line. Interpret slope: expected change in Y for +1 in X within data range. Avoid extrapolation. Intercept may be meaningless if $x = 0$ outside range. Residual diagnostics: residual vs fitted checks linearity + constant variance; Q-Q checks Normal errors for inference; residual vs order checks independence; Cook/leverage checks influence.

Fan shape ⇒ nonconstant variance; curve ⇒ wrong mean/form/add polynomial/transform; isolated point ⇒ influence/outlier check, not automatic deletion. Plot pattern Problem Response random band OK mean/var continue U curve nonlinearity transform fan heterosked. log/robust SE one far influence Cook/leverage

W5 PROBABILITY + BOX

Axioms: $0 \leq P(A) \leq 1$, $P(S) = 1$, disjoint additivity. Complements $P(A^c) = 1 - P(A)$. Inclusion-exclusion: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Conditional: $P(A|B) = P(A \cap B)/P(B)$; independence iff $P(A|B) = P(A)$ iff $P(A \cap B) = P(A)P(B)$; mutually exclusive nonempty events are not independent. $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Random variables: pmt/pdf/cdf. $E(aX + bY) = aE(X) + bE(Y)$ always; if independent, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. Bernoulli/Binomial: $X \sim \text{Bin}(n, p)$ counts successes; $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$; $E = np$, $\text{Var} = np(1 - p)$. Box model: draws from finite box. Sum of draws: expected sum = $n\mu_{\text{box}}$, SE sum = $\sqrt{n} \sigma_{\text{box}}/\sqrt{n}$.



Criterion	Likes	Warning
R^2	fit	always rises
adj R^2	fit+penalty	not CV
AIC	prediction-ish	can be complex
BIC	parsimonious	may underfit

W6 LOGISTIC + NONPARAM

Logistic regression: binary Y; $p_i = P(Y_i = 1 | x_i)$; logit link $\log(p_i/(1 - p_i)) = \eta_i = \beta_0 + \beta^T x_i$. $p_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}}$, odds = $\frac{p_i}{1 - p_i}$, OR = $e^{\beta^T x}$. Interpretation: β_j is change in log-odds per +1 x_j holding others fixed; e^{β_j} multiplies odds. Probability change depends on baseline p . 2x2 odds ratio: $a/b/c/d$ is a/b for x vs y exposure yes/no and c/d for event yes/no, OR = a/b if c/d is approximately the same. Logistic regression: > 1 high odds, not necessarily risk ratio. Prediction/classification: choose threshold using cost trade-off; confusion matrix, sensitivity, specificity, calibration matters. Do not infer causation from observational data. Nonparametric regression: smooth Y vs X without fixed linear form; bandwidth/span controls bias-variance: small span wiggly/low bias/high variance, large span smooth/high bias/low variance. Bootstrap: resample rows with replacement, recompute statistic; useful for median/mean/mode estimates where formula SE hard. Parametric bootstrap imposes model; nonparametric uses empirical distribution. HATPC ANSWER SCRIPTS Generic test paragraph: 'We test $H_0: \theta = \theta_0$ vs $H_A: \dots$. Assuming independent/random observations and the model-specific conditions, the statistic is $T = \dots$ Under H_0 , $T \sim \dots$ approximately. The p-value is $\dots$ since $\dots$ reject/fail to reject. In context, there is/is not sufficient evidence to $\dots$ CI paragraph: 'A 95% CI is estimate $\pm$ critical $\times$ SE = (L, U) . We are 95% confident the population parameter lies between L and U in context.' Link to test: two-sided 5% test rejects H_0 iff $|t|$ out of table 95% CI. Language traps: say 'fail to reject', not 'accept'; say 'evidence of association', not causal, unless randomized; say 'statistically significant at α ', not 'important'; say p-value conditional on H_0 .

Level	Use
90%	1.645 shorter CI
95%	1.96 standard
99%	2.576 wider CI

W7 CI + Z TEST

HATPC: H hypotheses; A assumptions; T statistic; P p-value; C conclusion in context. P-value = probability, assuming H_0 , of a statistic at least as extreme as observed in direction of H_A (vs H_0 true). Z test mean: known σ or large-count approximation: $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$. Proportion: $z = (\hat{p} - p_0)/\sqrt{p_0(1 - p_0)/n}$. Tail: $H_A: \theta = \theta_0$ upper, $<$ lower, $\neq$ two-sided. Choose direction before seeing data; post-hoc one-sided halves p-value invalid. Decision: reject if $p < \alpha$; otherwise fail to reject, never 'accept H_0 '. Practical importance needs effect size/CI, not only p-value. Type errors: Type I reject true H_0 , prob α ; Type II fail to reject false H_0 , power $1 - \beta$ increases with n , effect size, lower noise, larger α . Truth/Decision Reject Do not reject H_0 true Type I OK H_0 false Type II OK

W8 T TESTS

When t : σ unknown, use s ; statistic follows t , under Normal assumptions. Heavier tails than Normal; critical shrinks as df grows. $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$, $\bar{x} \pm t_{n-1} \cdot s/\sqrt{n}$. Paired t : same units before/after or matched pairs; analyze differences D , with one-sample t . Parameter μ_D , independence is between pairs, not within pair. Two-sample t : independent groups. Welch safer when variances/sample sizes differ; pooled t only if equal variances defensible. $t_{\text{Welch}} = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$. Assumption check: small n ; Q-Q/outliers; large n : CLT helps mean but not bad design. For paired sign-log simulation need null symmetry of differences. Sign test: use signs of paired differences when magnitude/Normality dubious; tests median/ μ systematic direction; less power if t assumptions OK. Recall focus: Test questions often require H/A/T/P/C plus assumption diagnosis, statistical p-value, and CI interpretation.

W9 CHI-SQUARE

Goodness-of-fit: one categorical variable vs specified probabilities. $E_i = n p_i$; $df = k - 1 - m$ if m parameters estimated from counts. Independence/homogeneity: two-way table; $E_{ij} = (\text{row}_i \text{col}_j)/n$; $df = (r - 1)(c - 1)$. $\chi^2 = \sum \frac{(O - E)^2}{E}$, cell contrib = $\frac{(O - E)^2}{E}$. Assumptions: counts; independent observations, mutually exclusive categories, expected counts mostly ≥ 5 (or all ≥ 1 and few below 5 depending rule). Conclusion: large χ^2 means observed far from expected; p-value is upper tail. Identify largest cell contribution and whether observed > expected. Simulation: generate counts with R , recompute statistic; $p = P(\chi^2_{df} \geq T_{\text{obs}}) / B$; if zero exceedances report $p < 1/B$ or $(0 + 1)/(B + 1)$, not $p = 0$. Chi-square EAO: compute expected counts, statistic, df, p-value bound, largest cell contribution, observed vs expected direction, validity of sampling design. Logistic EAO: convert log-odds to probability/odds, interpret odds ratio, compare groups, warn OR not risk ratio, load covariates fitted. Cue in question Likely method Do not forget before/after paired t differences 2 indep groups Welch/pooled variance issue counts by category χ^2 expected counts odds/prob binary responses logistic bootstrap expected counts odds/prob many predictors multiple lin assume rows resample rows

W10 MULTIPLE REGRESSION

Model: $\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \epsilon$, errors independent, mean 0, constant variance, approx Normal for inference. Coefficient β_j : adjusted association with Y for +1 in x_j , holding other predictors fixed. It is not automatically causal. Output: residual SE = $\sqrt{MSE}$; $R^2 = 1 - \text{SSE}/\text{SST}$; adjusted $R^2 = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)}$. t tests: $H_0: \beta_j = 0$ conditional on others. Overall F: $H_0: \beta_1 = \dots = \beta_p = 0$. Significant F with nonsig t can happen due joint correlation/collinearity. $F = \frac{(\text{SSE}_0 - \text{SSE})/(p - p_0)}{\text{SSE}/(n - p - 1)}$. Diagnostics: residual-fitted for form/variance; Q-Q for Normality; leverage high unusual X; Cook combines leverage+residual influence. Confounding: simple slope changes after adding covariate because covariate associated with both X and Y; adjusted interpretation is conditional. Lower better: BIC penalizes complexity more when $n > e^2$. Procedures: backward starts full and removes; forward starts null and adds; stepwise alternates. All are data-driven; p-values after selection are optimistic. Polynomial regression: include $x, x^2, \dots$ for curvature; keep hierarchy (if x^2 include x); interpret shape and region, not isolated coefficient. Model checks after selection: residual plots, influential points, collinearity, train/test or cross-validation if prediction. Avoid selecting solely by largest R^2 because R^2 never decreases when adding predictors.

W11 MODEL SELECTION

Nested models: reduced model is special case of full; compare with partial F. Do not use F for non-nested models. Selection goals: prediction vs explanation. Prediction prioritizes validation error; explanation prioritizes pre-specified covariates, interpretability, assumptions. $\text{AIC} = -2\ell + 2k$, $\text{BIC} = -2\ell + k \log n$. Lower better: BIC penalizes complexity more when $n > e^2$. Procedures: backward starts full and removes; forward starts null and adds; stepwise alternates. All are data-driven; p-values after selection are optimistic. Polynomial regression: include $x, x^2, \dots$ for curvature; keep hierarchy (if x^2 include x); interpret shape and region, not isolated coefficient. Model checks after selection: residual plots, influential points, collinearity, train/test or cross-validation if prediction. Avoid selecting solely by largest R^2 because R^2 never decreases when adding predictors.



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R^2	fit	always rises
adj R^2	fit+penalty	not CV
AIC	prediction-ish	can be complex
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W12 LOGISTIC + NONPARAM

Logistic regression: binary Y; $p_i = P(Y_i = 1 | x_i)$; logit link $\log(p_i/(1 - p_i)) = \eta_i = \beta_0 + \beta^T x_i$. $p_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}}$, odds = $\frac{p_i}{1 - p_i}$, OR = $e^{\beta^T x}$. Interpretation: β_j is change in log-odds per +1 x_j holding others fixed; e^{β_j} multiplies odds. Probability change depends on baseline p . 2x2 odds ratio: $a/b/c/d$ is a/b for x vs y exposure yes/no and c/d for event yes/no, OR = a/b if c/d is approximately the same. Logistic regression: > 1 high odds, not necessarily risk ratio. Prediction/classification: choose threshold using cost trade-off; confusion matrix, sensitivity, specificity, calibration matters. Do not infer causation from observational data. Nonparametric regression: smooth Y vs X without fixed linear form; bandwidth/span controls bias-variance: small span wiggly/low bias/high variance, large span smooth/high bias/low variance. Bootstrap: resample rows with replacement, recompute statistic; useful for median/mean/mode estimates where formula SE hard. Parametric bootstrap imposes model; nonparametric uses empirical distribution. HATPC ANSWER SCRIPTS Generic test paragraph: 'We test $H_0: \theta = \theta_0$ vs $H_A: \dots$. Assuming independent/random observations and the model-specific conditions, the statistic is $T = \dots$ Under H_0 , $T \sim \dots$ approximately. The p-value is $\dots$ since $\dots$ reject/fail to reject. In context, there is/is not sufficient evidence to $\dots$ CI paragraph: 'A 95% CI is estimate $\pm$ critical $\times$ SE = (L, U) . We are 95% confident the population parameter lies between L and U in context.' Link to test: two-sided 5% test rejects H_0 iff $|t|$ out of table 95% CI. Language traps: say 'fail to reject', not 'accept'; say 'evidence of association', not causal, unless randomized; say 'statistically significant at α ', not 'important'; say p-value conditional on H_0 .

Method	Parameter	Cue assumption
one z/t test	μ	sample Normal/large n
paired t	μ_D	paired diff Normal/large n
Welch t	$\mu_1 - \mu_2$	indep groups
prop z	p	large counts
χ^2	prob/assoc	expected counts
regression β_j	β_j	linear model

R COMMAND MAP

Normal: `pnorm(x, mean, sd)` left prob; `1 - pnorm(x, ...)` upper; `pnorm(b, ..., pnorm(a, ...))` interval; `qnorm(p, ...)` cutoff; `dnorm` density only. Tests: `t.test(x, mu0, one.sample)`; `t.test(after, before, paired=TRUE)` paired; `t.test(y, group, var.equal=FALSE)` Welch; `prop.test(x, n, p0, prop)` proportion approx; `chisq.test(tab)` table; `chisq.test(obs, pprob)` GOF. Regression: `lm(y ~ x)`, `lm(y ~ x1 + x2)`, `lm(y ~ x1(x^2))`; summary(fit); `coeff(fit)`; `anova`; `resid`, `null`, `fitted`; `plot(fit)` diagnostics; `cooks.distance(fit)` influence. Logistic/smooth: `glm(y ~ x1 + x2, family=binomial)`; odds ratio `exp(coef(fit))`; fitted prob `fitted(fit, type="response")`; nonparam smooth `geom_smooth(method="loess")` or `loess(x, y)`. R output reading: Estimate is coefficient; Std. Error is SE of estimate; value of z value is estimate; $|z| > 1.96$ two-sided p-value; residual df = $n - p - 1$.

DISTRIBUTION BANK

Dist	Use	Mean	Var
Bern(p)	one success	p	$p(1 - p)$
Bin(n, p)	count	np	$np(1 - p)$
Geom(p)	measure	$1/p$	$(1 - p)/p^2$
Norm(μ, σ^2)	mean, σ unk	μ	σ^2
χ^2_k	sum sqs std resd	k	$2k$

Standardisation: if $X \sim N(\mu, \sigma^2)$ then $Z = (X - \mu)/\sigma \sim N(0, 1)$. If X_i iid, X has mean μ and SE $\sigma/\sqrt{n}$; use $s/\sqrt{n}$ if σ unknown. Sampling without replacement: draws dependent; finite-pop correction may be needed. STA: count enumeration questions ask exact small-sample distribution instead. For sample proportion under $H_0: p = p_0$, use SE with p_0 in test; for CI use $\hat{p}$ in SE.

PAST-EXAM PATTERNS

Graph/numeric MCQ: add/subtract constant; kg to lb (167, 165, 5, 59) not 167, 165, 5, 59.2. Boxplot percentile; correlation under transformations; normal R code. First rule: identify type of transformation before calculating. Inference EAO: HATPC assumptions, validity bound from central table, contextual conclusion, then CI or simulation comparison. Recalls emphasize t , two-sample t , F, chi-square, bootstrap/simulation. Regression EAO: write fitted equation, interpret coefficient, state diagnostic assumptions, compute residual, explain R^2 /F-test, distinguish adjusted association from causation. Chi-square EAO: compute expected counts, statistic, df, p-value bound, largest cell contribution, observed vs expected direction, validity of sampling design. Logistic EAO: convert log-odds to probability/odds, interpret odds ratio, compare groups, warn OR not risk ratio, load covariates fitted. Cue in question Likely method Do not forget before/after paired t differences 2 indep groups Welch/pooled variance issue counts by category χ^2 expected counts odds/prob binary responses logistic bootstrap expected counts odds/prob many predictors multiple lin assume rows resample rows

COMMON PITFALLS

Density: histogram y-axis may be density, not count; area = proportion. Normal R: third argument is σ , not variance. `pnorm(167, 165, 5, 59)` not 167, 165, 5, 59.2. Correlation: slope and correlation are not the same; changing units of one variable changes slope but not $|r|$; negative scaling flips sign. CLT: sample mean approx Normal; original data need not become Normal; sample size does not remove bias from bad sampling. P-value: high p-value means count not surprising under H_0 , not proof H_0 true; low p-value does not mean effect size. Multiple testing: 20 true nulls at $\alpha = .05$ give expected false positives 20 $\times$.05 = 1, family-wise chance at least one false positive = .99. Pooled t : do not choose pooled merely because test p-value $> .05$; with unequal n and very different s^2 , Welch is safer. Model selection: selected model p-values understate uncertainty; report selection criterion and validate. Outliers: delete needs scientific/statistical justification; influential point can change slope, SE, and p-values. MINI EXAMPLES CI width: if n quadruples, SE halves, CI width halves; 99% wider than 95%, same centre if same data. One/two-sided: if two-sided $p = .08$ and observed effect in positive direction, preplanned upper one-sided $p = .04$; if chosen after seeing sign, invalid. CUE: sample mean approx Normal; original data need not become Normal; sample size does not remove bias from bad sampling. P-value: high p-value means count not surprising under H_0 , not proof H_0 true; low p-value does not mean effect size. Multiple testing: 20 true nulls at $\alpha = .05$ give expected false positives 20 $\times$.05 = 1, family-wise chance at least one false positive = .99. Pooled t : do not choose pooled merely because test p-value $> .05$; with unequal n and very different s^2 , Welch is safer. 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